

Software Design Document (SDD)

Automated Student Financial Clearance System

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1. INTRODUCTION

1.1 Purpose

This Software Design Document (SDD) describes the architecture and system design of the Automated Student Financial Clearance System. The intended audience includes software developers, the UNESWA ICT department, and the University Finance Office.

1.2 Scope

The system provides a digital platform for students to submit financial documentation and receive clearance. It integrates with the existing UNESWA Moodle platform to ensure a seamless experience. Key goals include reducing manual paperwork, preventing fraud, and supporting 5,000 concurrent users.

1.3 Overview

This document provides a high-level view of the system architecture followed by detailed data and component designs.

1.4 Reference Material

- Software Requirements Specification (SRS) for Automated Student Financial Clearance System
- IEEE-Std 1016-2009
- Eswatini Data Protection Act

1.5 Definitions and Acronyms

SRS: Software Requirements Specification

UNESWA: University of Eswatini

PoP: Proof of Payment

SDD: Software Design Document

2. SYSTEM OVERVIEW

The system automates the financial clearance workflow. Students upload IDs and Proof of Payment via Moodle. The system uses hash-based fraud detection and automated verification to update clearance status in the university database.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

The system uses a Client-Server architecture. The Moodle platform acts as the client portal, while the backend serves business logic and manages the database repositories (MySQL/Postgres/MongoDB).

3.2 Decomposition Description

The system is decomposed into the following modules: Authentication Module (Moodle SSO), Document Handling Subsystem, Verification Engine, and Database Sync Service.

3.3 Design Rationale

Modular design was chosen to ensure that the Verification Engine can be scaled independently during peak registration periods to handle the required 5,000 concurrent logins.

4. DATA DESIGN

4.1 Data Description

Data is structured to secure personal financial records. Entities include Student Credentials (masked), Document Metadata (hash strings), and Clearance Status logs.

4.2 Data Dictionary

Entity	Type	Description
StudentID	String	Unique student identifier from Moodle
DocHash	String	SHA-256 hash of uploaded file for fraud check
ClearanceStatus	Boolean	Indicates if student is cleared for registration

5. COMPONENT DESIGN

Document Verification Pseudocode:

```
IF upload.isComplete() THEN
  hash = computeSHA256(file)
  IF db.findHash(hash) THEN
    REJECT "Duplicate Document"
  ELSE
```

```
db.updateStatus(studentID, 'Pending')
END IF
END IF
```

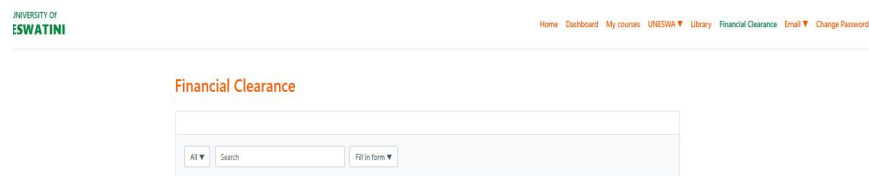
6. HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The interface is integrated into Moodle using a single-page upload portal with masked password entry.

6.2 Screen Images

UNESWA Portal



Student Clearance Form

The screenshot displays the 'Student Financial Clearance Gateway' form, which includes the following sections:

- 1. Academic & Personal Identity**: Contains fields for 'Student ID Number' (with example 'e.g., 20261024') and 'Full Name' (with sub-field 'Surname First Name').
- 2. Clearance Scope**: Includes a 'Purpose of Clearance' dropdown menu (selected: 'Tuition (Registration / Results)') and a 'Sponsorship Class' dropdown menu (selected: 'Government Sponsored').
- 3. Document Verification Vault**: Features two upload areas: 'Official Bank Deposit Slip / Statement' and 'Government Scholarship Award Letter', both with instructions to 'Drag & drop or Click to choose PDF/Image'.

At the bottom of the form, there are two buttons: 'Cancel Process' and 'Submit to Finance System'.

6.3 Screen Objects and Actions

- Upload Button: Triggers file selection and validation.
- Status Indicator: Shows 'Cleared', 'Failed', or 'Pending'.

7. REQUIREMENTS MATRIX

SRS Requirement	System Component
3.2.1 Document Submission	Submission Subsystem
3.2.2 Security (Fraud)	Verification Subsystem
3.3 Performance (5k Users)	Database/Load Balancer

8. APPENDICES

N/A